DESIGN AND IMPLEMENTATION OF A WEB SYSTEM FOR SEARCHING AND MANAGING ACADEMIC COURSES

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INTRODUCTION

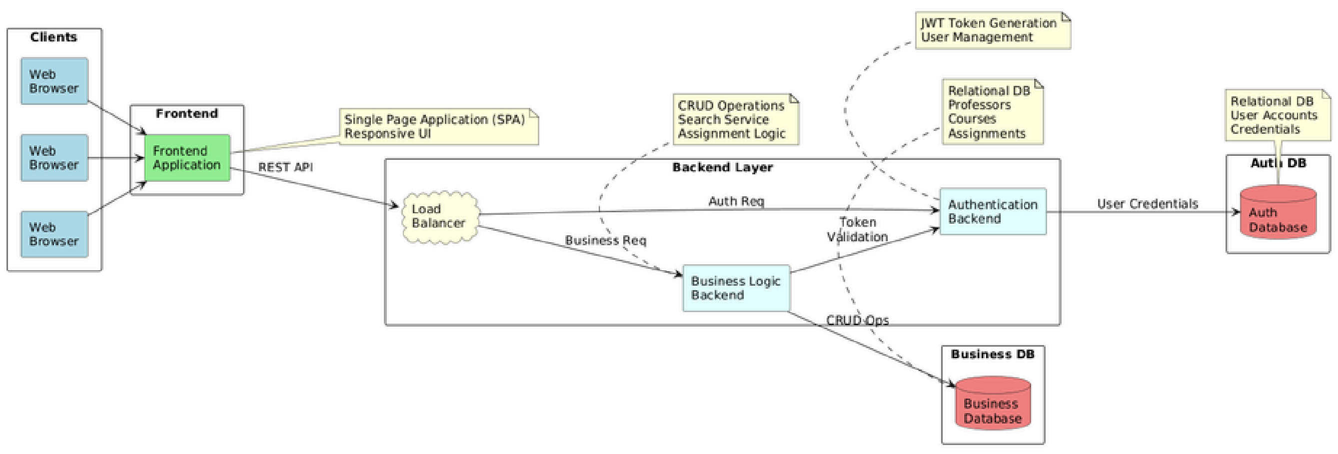
Universities face difficulties in managing and accessing academic course information. Students and teachers often struggle to find who teaches what, due to scattered data across different platforms. This project proposes a web-based system that centralizes academic information and facilitates efficient search and management.

GOAL

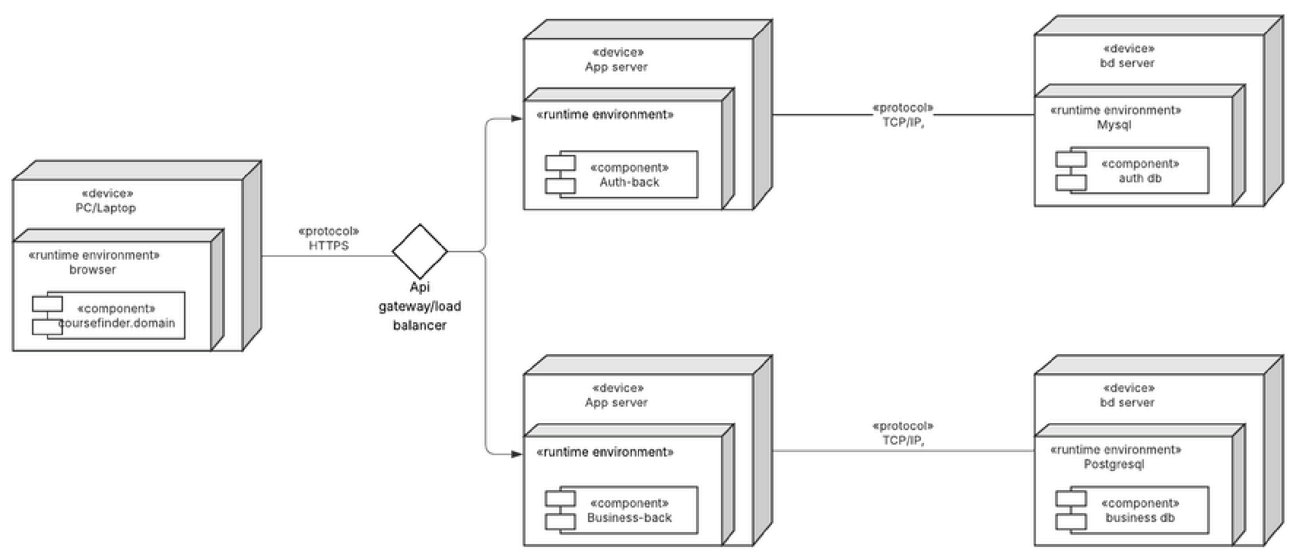
To design and implement a modular web system that allows efficient course and professor management within a single institution. The final product provides two user roles: general users for course search and an administrator for data management.

METHODS

The system was built using a microservices-based architecture composed of two independent backends: a Java authentication service managing login and token validation, and a Python business service handling all course and professor operations.



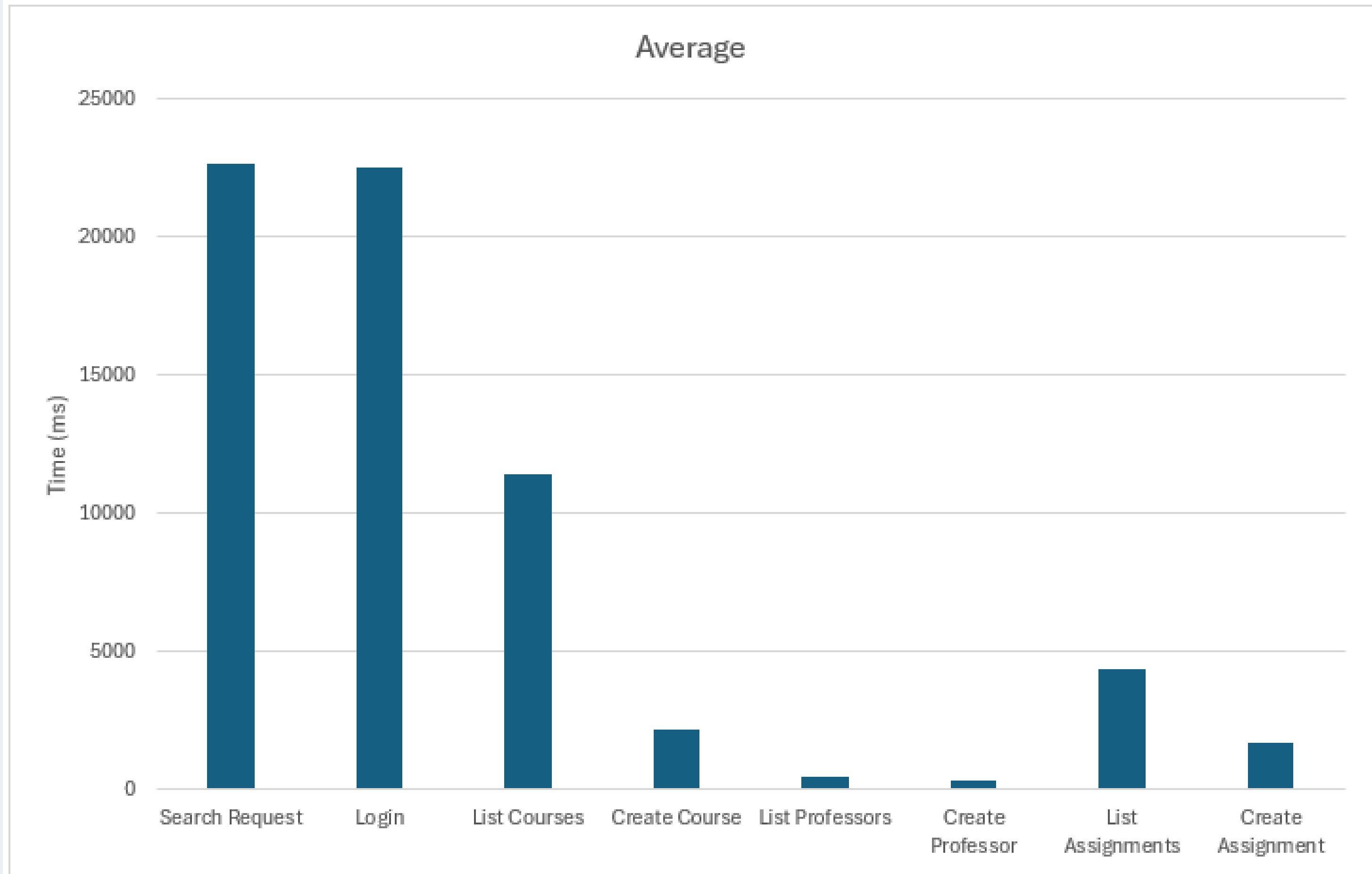
Both services expose REST APIs and are fully containerized and the overall structure follows a layered design—presentation, business logic, and data management—deployed as separate components in the system's architecture. These elements are represented in the deployment and design diagrams, which outline how the services communicate.

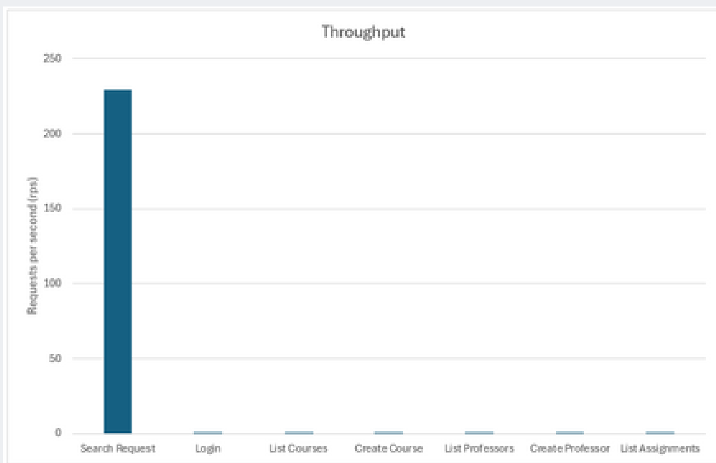


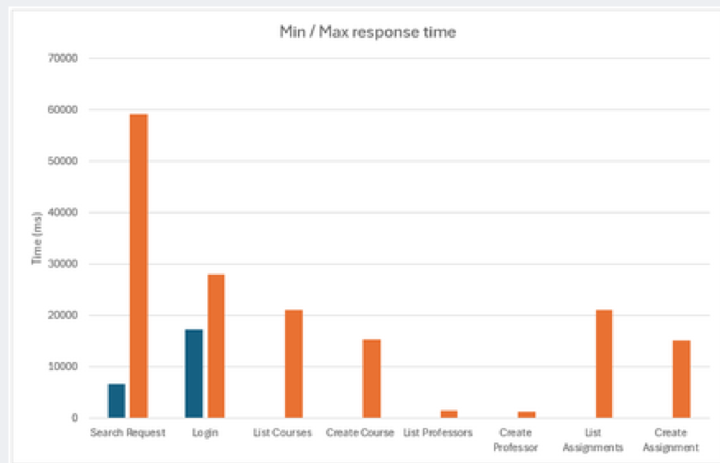
The architecture also supports bidirectional search logic and secure authentication flows, both of which are reflected in the system's design diagrams.

RESULTS

The system successfully met all defined use cases and passed both unit and acceptance tests for course and instructor registration, search, and assignment.







The microservices architecture performed reliably under moderate load, though stress testing revealed a limitation in the search endpoint under high concurrency due to hardware constraints. Overall, the results validate the system's design and highlight the need for future work focused on infrastructure scaling and performance optimization. Future enhancements should also expand the system's functionality such as enrollment features and personalized tools to continue aligning with the original objective of improving course discovery and academic planning.

BIBLIOGRAPHY

[1] H. DRACHSLER, K. VERBERT, O. C. SANTOS, AND N. MANOUSELIS, "PANORAMA OF RECOMMENDER SYSTEMS TO SUPPORT LEARNING," IN RECOMMENDER SYSTEMS HANDBOOK, F. RICCI, L. ROKACH, B. SHAPIRA, AND P. B. KANTOR, EDS. SPRINGER, 2015, PP. 421–451.

[2] J. LU, "A SURVEY OF ONLINE COURSE RECOMMENDATION TECHNIQUES," OPEN JOURNAL OF APPLIED SCIENCES, VOL. 12, PP. 134–154, 2022.